Table 1: Massive-STEPS cities. *Queries*: test next-visit queries kept by the STHGCN protocol (target POI and user seen in training). *Revisit*: share of check-ins to an already visited place, over all check-ins and over the scored queries. γ: novelty exponent of $P(\text{new}) \propto S^{-\gamma}$.

City	Users	Check-ins	POIs	Queries	Revisit		γ
					all	test	
Source (training)							
Istanbul	23,700	544,471	42,795	41,150	0.40	0.50	0.33
Petaling Jaya	14,308	506,430	53,650	33,889	0.37	0.44	0.24
Jakarta	8,336	412,100	71,150	24,402	0.41	0.53	0.22
New York	6,929	272,368	45,804	16,766	0.37	0.45	0.27
Kuwait City	9,628	232,706	13,455	17,753	0.45	0.56	0.46
São Paulo	5,822	256,824	34,881	16,523	0.51	0.62	0.31
Unseen (zero-shot)							
Tokyo	764	13,839	2,982	946	0.42	0.60	0.34
Shanghai	296	10,491	3,610	530	0.36	0.53	0.17
Palembang	267	14,467	4,117	774	0.44	0.67	0.29
Melbourne	646	22,050	6,901	1,239	0.35	0.52	0.19
Sydney	740	29,900	8,189	1,667	0.41	0.55	0.48
Tangerang	1,437	45,521	9,975	2,787	0.51	0.68	0.31
Moscow	3,993	105,620	13,994	7,288	0.43	0.54	0.28
Bandung	3,377	161,284	26,559	9,913	0.43	0.55	0.28

GEMMA – figures and tables (preview)

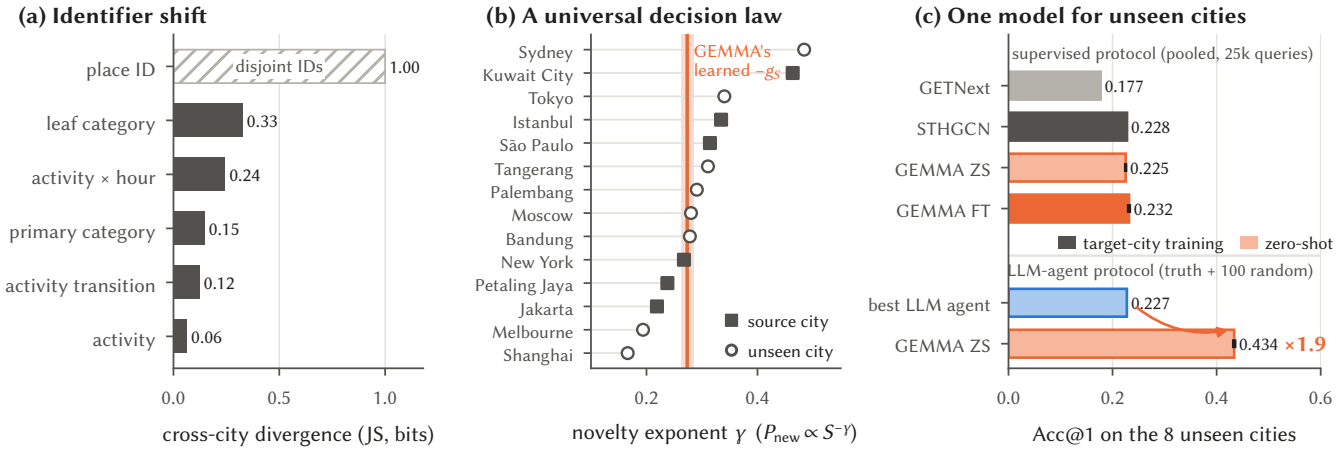


Figure 1: What transfers across cities. (a) Place identifiers are disjoint between cities (cross-city Jensen–Shannon divergence = 1 bit); the more abstract the description of a visit, the more cities share it. (b) The novelty exponent γ of the exploration law $P(\text{new}) \propto S^{-\gamma}$ in the 14 evaluated cities; the orange band is the gate coefficient $-g_S$ that GEMMA learns on the six source cities without ever being given γ. (c) Accuracy on the eight cities GEMMA never saw: zero-shot GEMMA is within 0.3 points of STHGCN, which is trained in each city, and nearly doubles the best LLM agent; light fine-tuning on the training split exceeds STHGCN.

Table 2: Next-POI prediction on the eight cities GEMMA never saw during training (STHGCN protocol; baselines from Massive-STEPS). *Target train*: the model is trained on that city's training split. GEMMA zero-shot uses no data from the city except each test user's own past check-ins; fine-tuned GEMMA uses the training split only. Mean of 3 seeds. Best in **bold**, second underlined. A@1 = Acc@1, N@5 = NDCG@5; pooled = weighted by the number of test queries.

	Target	Tokyo		Shanghai		Palembang		Melbourne		Sydney		Tangerang		Moscow		Bandung		Average		
Model	train	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	$A@1_{\text{pooled}}$
FPMC	✓	.176	.239	.084	.120	.102	.136	.062	.107	.075	.131	.104	.166	.059	.094	.048	.083	.089	.135	.067
RNN	✓	.133	.197	.055	.090	.049	.085	.059	.083	.080	.125	.087	.135	.075	.122	.062	.099	.075	.117	.072
LSTPM	✓	.225	.315	.099	.149	.114	.175	.091	.150	.141	.206	.154	.237	.151	.229	.110	.179	.136	.205	.132
DeepMove	✓	.201	.288	.085	.128	.084	.139	.083	.134	.129	.188	.145	.219	.143	.217	.107	.172	.122	.186	.124
Flashback	✓	.170	.237	.096	.132	.135	.198	.079	.115	.127	.174	.185	.262	.121	.180	.103	.155	.127	.182	.121
Graph-Flashback	✓	.180	.259	.111	.151	.148	.208	.093	.143	.135	.188	.200	.292	.125	.188	.091	.146	.135	.197	.121
LoTNext	✓	.191	.258	.093	.139	.140	.208	.085	.133	.123	.176	.126	.190	.089	.145	.076	.125	.115	.172	.096
GETNext	✓	.180	.275	.115	.177	.158	.239	.100	.179	.181	.266	.224	.302	.175	.260	.179	.247	.164	.243	.177
STHGCN	✓	.250	.350	**.193**	.264	.246	.341	**.168**	.247	**.227**	**.307**	.293	.400	.223	.308	**.219**	.302	.227	.315	.228
GNPR-SID	✓	.114	.126	.060	.067	.094	.103	.049	.054	.105	.112	.165	.177	.116	.128	.126	.136	.104	.113	.119
UniMove	✓	.032	.072	.000	.029	.009	.035	.008	.037	.015	.059	.001	.029	.009	.030	.007	.033	.010	.040	.008
GEMMA zero-shot	×	.261	.366	.170	.267	**.248**	.341	.167	.253	.204	.287	.319	.435	.219	.315	.212	.296	.225	.320	.225
GEMMA fine-tuned	✓	**.271**	**.386**	.184	**.279**	.245	**.348**	.165	**.263**	.221	**.307**	**.332**	**.448**	**.225**	**.325**	.216	**.307**	**.232**	**.333**	**.232**

Table 3: Next-POI prediction on the six source cities GEMMA was trained on (in-domain; fine-tuned GEMMA, 3 seeds).

	Target	Istanbul		P. Jaya		Jakarta		New York		Kuwait C.		São Paulo		Average		
Model	train	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	A@1	N@5	$A@1_{\text{pooled}}$
FPMC	✓	.026	.050	.026	.057	.029	.058	.032	.061	.021	.054	.030	.055	.027	.056	.027
RNN	✓	.077	.130	.064	.107	.049	.083	.061	.092	.087	.146	.097	.147	.072	.118	.071
LSTPM	✓	.142	.217	.099	.163	.099	.157	.099	.155	.180	.275	.158	.243	.130	.202	.127
DeepMove	✓	.150	.228	.112	.175	.103	.160	.097	.149	.179	.274	.160	.240	.134	.204	.132
Flashback	✓	.137	.200	.072	.119	.113	.166	.086	.123	.150	.227	.159	.222	.120	.176	.117
Graph-Flashback	✓	.137	.202	.072	.119	.072	.121	.059	.093	.115	.181	.096	.156	.092	.145	.096
LoTNext	✓	.105	.168	.055	.097	.044	.078	.061	.099	.132	.209	.105	.166	.084	.136	.082
GETNext	✓	.146	.210	.139	.200	.155	.209	.134	.202	.175	.251	.202	.286	.159	.226	.154
STHGCN	✓	**.241**	**.318**	**.174**	**.241**	.197	.270	.146	.207	**.225**	**.314**	**.250**	.344	**.206**	.282	**.207**
GNPR-SID	✓	.115	.127	.090	.098	.102	.107	.074	.080	.162	.178	.165	.176	.118	.128	.114
UniMove	✓	.015	.038	.008	.034	.004	.020	.004	.016	.023	.073	.002	.009	.009	.032	.010
GEMMA fine-tuned	✓	.198	.300	.156	.239	**.216**	**.302**	**.155**	**.233**	.204	.307	.247	**.359**	.196	**.290**	.193

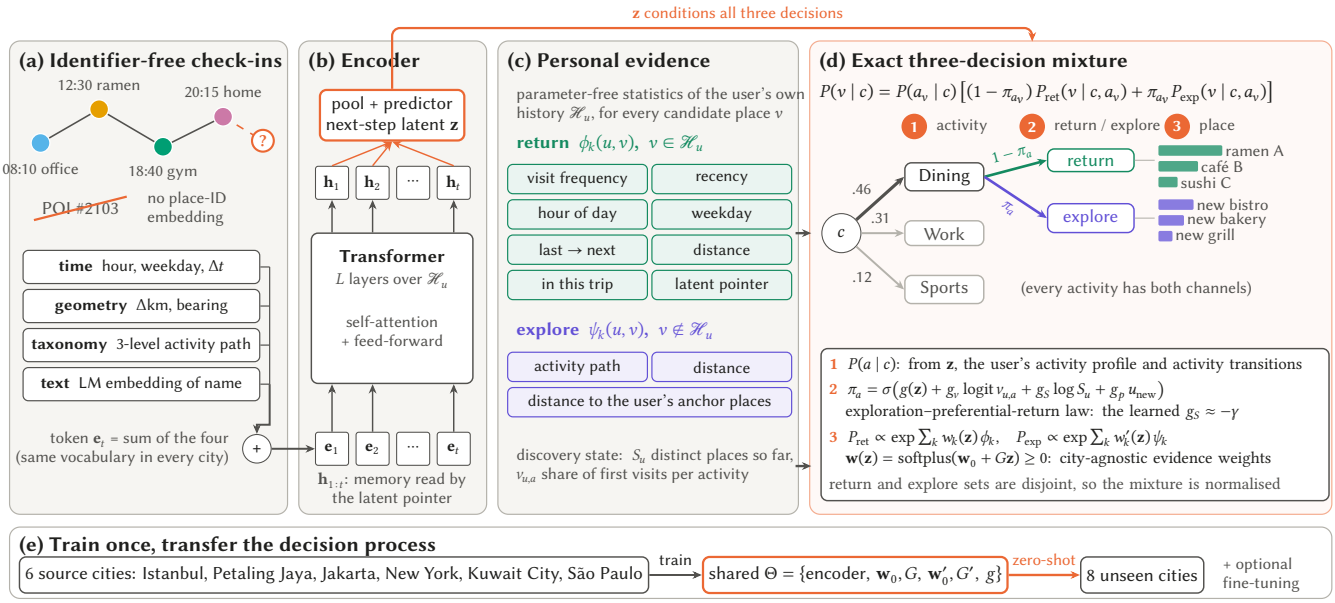


Figure 2: GEMMA. (a) Check-ins are tokens built from time, relative geometry, the activity taxonomy and a language embedding of the place, never from a place identifier. (b) A Transformer encodes the user's history into states $\mathbf{h}_{1:t}$ and a next-step latent $\mathbf{z}$. (c) Parameter-free statistics of the user's own history score every candidate place. (d) The next-place distribution factorises exactly into three decisions: the activity, a gate between returning and exploring that follows the exploration–preferential-return law, and the place, scored by non-negative, context-modulated evidence weights. (e) All parameters act on statistics, so the model trained on six cities is applied unchanged to unseen cities.

Table 4: Zero-shot comparison with LLM agents under the LLM-Move protocol of Massive-STEPS: candidates are the true next place plus 100 random places of the city, history = the user's last 15 trajectories, Acc@1. LLM rows are the published numbers (200 users); GEMMA is the mean over 3 seeds × 5 user samples of 200 users. GEMMA uses no LLM at inference.

Framework	Model	Tokyo	Shanghai	Palembang	Melbourne	Sydney	Tangerang	Moscow	Bandung	Avg.
LLM-Mob	Gemini 2 Flash	.140	.055	.135	.060	.060	.155	.130	.105	.105
	Claude Haiku 4.5	.180	.070	.135	.070	.075	.155	.130	.115	.116
	Qwen 2.5 7B	.110	.040	.075	.030	.035	.095	.090	.060	.067
	Llama 3.1 8B	.005	.005	.005	.010	.020	.020	.030	.010	.013
	Gemma 2 9B	.145	.050	.095	.055	.030	.145	.100	.070	.086
LLM-ZS	Gemini 2 Flash	.160	.065	.130	.065	.060	.145	.125	.095	.106
	Claude Haiku 4.5	.205	.075	.165	.070	.070	.175	.130	.115	.126
	Qwen 2.5 7B	.120	.045	.050	.040	.045	.100	.080	.055	.067
	Llama 3.1 8B	.110	.060	.070	.040	.040	.080	.080	.045	.066
	Gemma 2 9B	.110	.050	.060	.050	.045	.100	.080	.065	.070
LLM-Move	Gemini 2 Flash	.250	.170	.260	.225	.230	.200	.220	.225	.223
	Claude Haiku 4.5	.255	.100	.175	.175	.235	.220	.230	.205	.199
	Qwen 2.5 7B	.250	.095	.130	.110	.125	.175	.230	.100	.152
	Llama 3.1 8B	.030	.020	.010	.040	.055	.000	.005	.030	.024
	Gemma 2 9B	.130	.105	.210	.115	.125	.125	.110	.175	.137
GEMMA	zero-shot, no LLM	**.477**	**.305**	**.406**	**.326**	**.337**	**.592**	**.521**	**.507**	**.434**

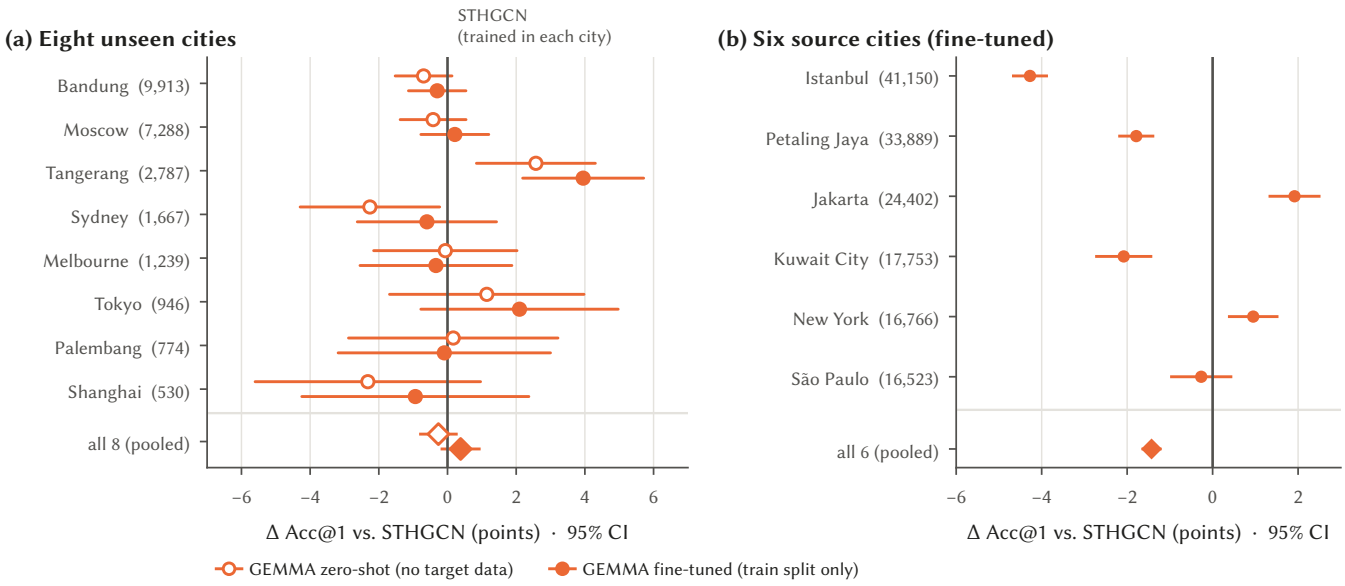


Figure 3: Per-city difference to STHGCN (Acc@1 points, 95% CI from test sampling and seed spread; number of test queries in brackets). (a) On unseen cities, zero-shot GEMMA (hollow) is statistically indistinguishable from STHGCN in 6 of 8 cities (better in Tangerang, worse in Sydney); fine-tuned GEMMA (solid) is indistinguishable in 7 and significantly better in Tangerang, with no significant loss. (b) On the six source cities STHGCN keeps an in-domain advantage in Istanbul, Petaling Jaya and Kuwait City; GEMMA is better in Jakarta and New York.

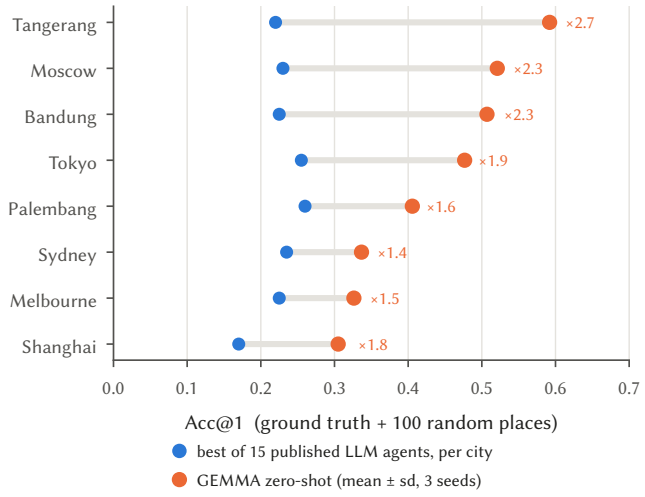


Figure 4: Zero-shot Gemma vs. LLM agents under the LLM-Move protocol. Blue: the best of the 15 published LLM agents in each city (an oracle choice of framework and model); orange: Gemma (3 seeds), which runs no LLM.

Table 5: Aggregate results on the eight unseen cities. Fine-tuned GEMMA is shown with the test-time history drawn from the training split only (as STHGCN) and from train+validation.

	Acc@1		NDCG@5
Model	macro	pooled	macro
GETNext	0.1640	0.1771	0.2431
STHGCN	0.2274	0.2278	0.3149
GEMMA zero-shot	0.2250	0.2252	0.3200
GEMMA FT (train)	0.2324	0.2316	0.3330
GEMMA FT (train+val)	0.2348	0.2334	0.3359

Table 6: Ablations: change of zero-shot macro Acc@1 (points, eight unseen cities).

Variant	ΔAcc@1
Learned evidence weights → hand-set	-2.5
no world-model objective	+0.1
no JEPA objective	+0.0
no backbone pretraining	-0.1
flat activity taxonomy	+0.1
no latent pointer	-0.1

Table 7: Learned parameters (at $\mathbf{z} = 0$). Source: trained on six cities, mean ± sd over 3 seeds. Fine-tuned: range over the eight unseen cities × 3 seeds after fine-tuning. † needs the target-city log, inactive in zero-shot.

Parameter	Source-trained	Fine-tuned (range)
Return evidence $\mathbf{w}_0$		
frequency	0.63 ± 0.04	0.46 – 0.75
in this trip	0.07 ± 0.03	0.04 – 0.23
is last place	-1.69 ± 0.02	-2.78 – -1.47
last→next	1.02 ± 0.02	0.89 – 1.07
hour of day	0.44 ± 0.01	0.39 – 0.52
weekday	0.41 ± 0.01	0.36 – 0.46
recency	1.32 ± 0.05	1.07 – 1.46
distance	0.39 ± 0.03	0.35 – 0.74
latent pointer	-0.01 ± 0.01	-0.04 – 0.04
Explore evidence $\mathbf{w}'_0$		
activity path	0.37 ± 0.01	0.28 – 0.40
distance	0.97 ± 0.08	0.87 – 1.69
anchor distance	0.56 ± 0.01	0.48 – 0.80
co-visit†	1.00 ± 0.00	0.92 – 1.07
popularity†	0.30 ± 0.00	0.13 – 0.27
city hour profile†	0.30 ± 0.00	0.19 – 0.30
Gate		
g_v novelty rate	0.82 ± 0.01	0.73 – 0.81
g_S log #places	-0.27 ± 0.01	-0.30 – -0.23
g_P new prefix	0.86 ± 0.02	0.65 – 1.00

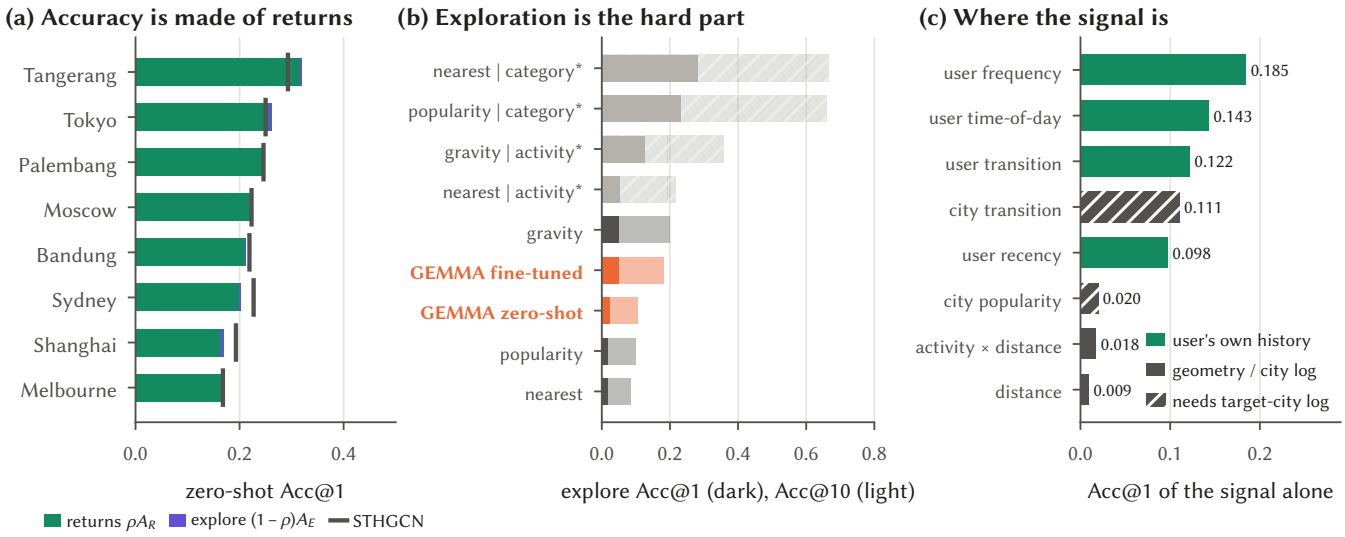


Figure 5: Anatomy of the prediction. (a) Zero-shot Acc@1 split into returns (ρA_R) and explorations ($(1-\rho)A_E$); almost all hits are returns. (b) Ranking a *new* place, given that the user explores (mean over the 8 unseen cities): GEMMA beats distance and popularity rankers, but oracles that know the true activity or category (*, hatched) show how much remains. (c) Single signals ranked alone: the user's own frequency, time-of-day and transitions carry most of the information; hatched signals need a log of the target city.

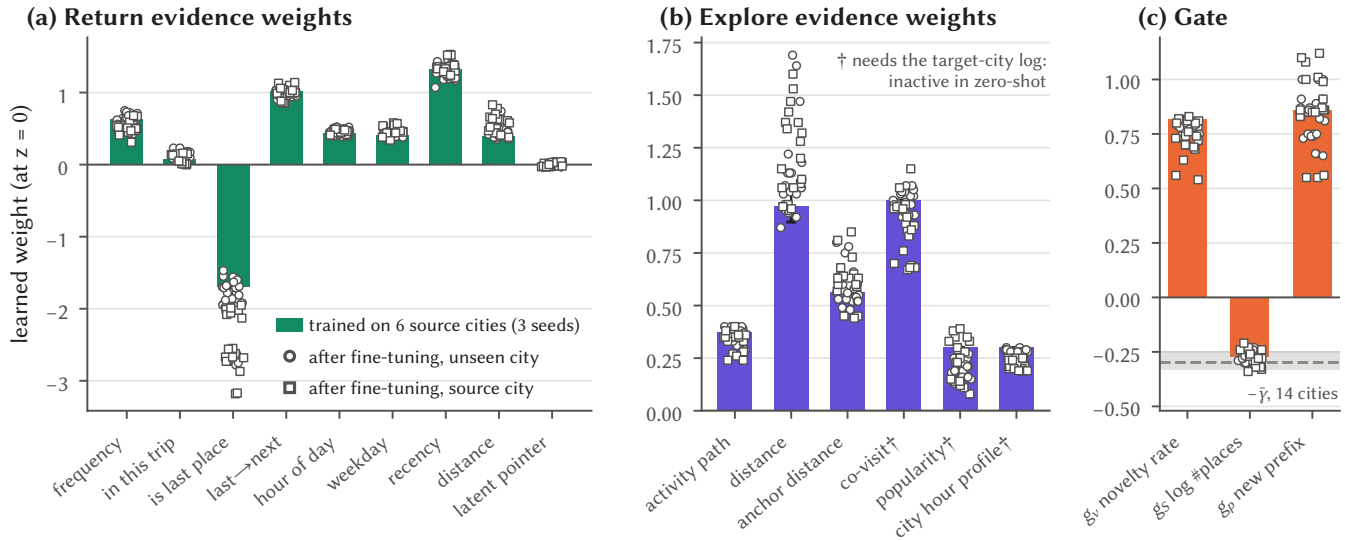


Figure 6: Learned parameters. Bars: values learned on the six source cities (mean ± sd over 3 seeds). Markers: values after fine-tuning in each unseen (circles) or source (squares) city. Fine-tuning changes the return weights by a median of 8% (relative L_1); only the largest cities move further, mostly through the is-last penalty. (c) The gate's coefficient on $\log S_u$ falls inside the range of $-\gamma$ measured in the 14 evaluated cities (grey band: inter-quartile range).

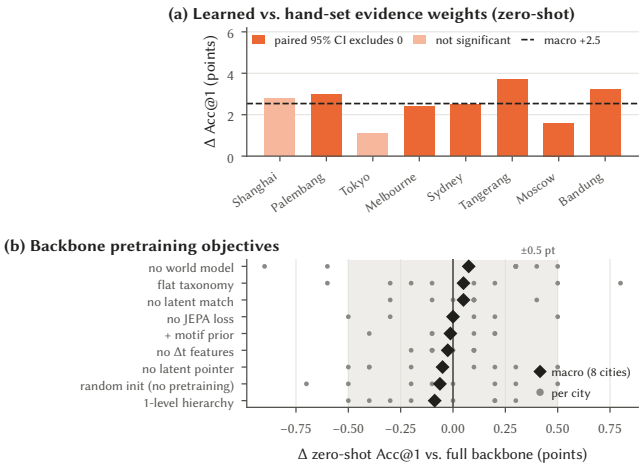


Figure 7: Ablations (zero-shot). (a) Learning the evidence weights on source cities instead of setting them by hand adds 2.5 points (macro over 8 unseen cities). (b) Removing representation-learning objectives from the backbone changes accuracy by less than 0.1 point on average: the transfer does not come from pretraining the encoder.

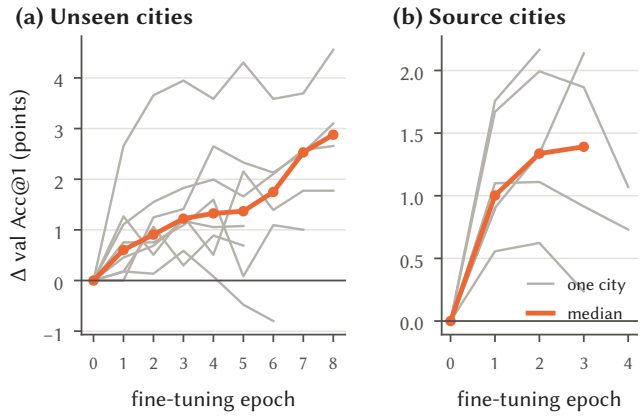


Figure 8: Fine-tuning dynamics. Validation Acc@1 relative to the transferred model (epoch 0). Gains are small and arrive within the first epochs.